Does supplying documents eliminate hallucination?

Retrieval

Was the evidence found?

Correctness

Is the answer right?

Support

Does the source back each claim?

CS420 Seminar S07 · RAG and Evidence Support · Phan Anh Khoa (24125059) · Bùi Trọng Thành (24125043)

Where the knowledge lives

Parametric memory

Learned model weights

Non-parametric memory

An external store, searched when asked

Approach	Knowledge in	At answer time	To update
Fine-tuning	Weights	Use adapted model	Train again
Full-document prompt	Input	Read documents	Supply new text
RAG	External store	Retrieve top-k, generate	Update index

Team synthesis based on Lewis et al., NeurIPS 2020

Each pipeline stage can fail differently

Ingest
PDF text or table layout is lost

→

Chunk
A condition is split from its claim

→

Index
Outdated chunks remain

→

Retrieve
Passage is missing or wrong

→

Prompt
Metadata is lost or cut off

→

Generate
Claim goes beyond evidence

Then verify

Keep each chunk’s ID and page, and check every claim against the passage it cites.

Project pipeline informed by Lewis et al., NeurIPS 2020

RAG-Sequence and RAG-Token

Sequence

$$p(y|x) \approx \Sigma z \; p\eta(z|x) \cdot p\theta(y|x,z)$$

One document for the whole output

z1: 0.6 × 0.7 = 0.42
z2: 0.4 × 0.2 = 0.08
0.42 + 0.08 = **0.50**

Token

$$p(y|x) \approx \Pi i \; \Sigma z \; p\eta(z|x) \cdot p\theta(y i|x,z,y<i)$$

Documents mixed at each token

y1: 0.6 × 0.875 + 0.4 × 0.5 = 0.725
y2: 0.6 × 0.8 + 0.4 × 0.4 = 0.64
0.725 × 0.64 = **0.464**

x question · **y** output · **z** document in top-k · **η** retriever · **θ** generator · probabilities in [0, 1]

Our API demo concatenates chunks, so it implements neither formula.

Equations: Lewis et al., NeurIPS 2020 · Numbers are hypothetical, not from Gemini

One open paper, twenty checkable questions

7

pages

4 main text, 3 mostly references

20

questions

Each with a reference answer and gold spans

CC BY

4.0 article

The abess software itself is GPL-3

Corpus: abess, a best-subset selection library (JMLR 2022).

Same chunks and scoring, different vectors

TF-IDF

Sparse word and phrase weights
Unigrams and bigrams
Rewards exact term matches

all-MiniLM-L6-v2

Dense 384-dimensional embeddings
Pretrained, not adapted to the corpus
Captures broader similarity

Chunk size	Overlap	Top-k	Scoring
96 or 192 tokens	24 tokens	1, 3, 5	Exact cosine

Pedregosa et al., 2011; all-MiniLM-L6-v2 model card · 81 or 36 chunks

Recall measures finding, not answering

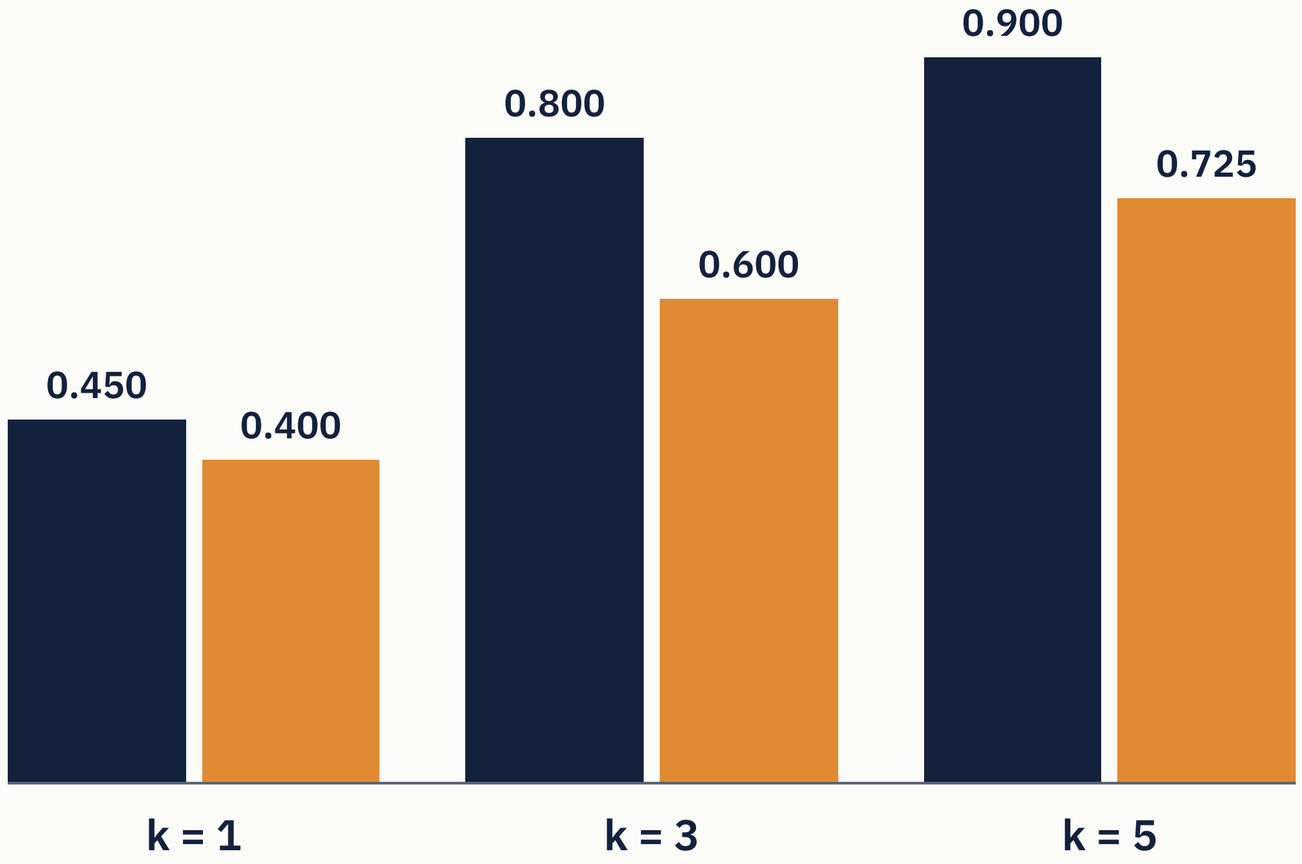
$$\text{Recall@k} = |\ \text{relevant} \cap \text{top-k}\ | \ /\ |\ \text{relevant}\ |$$

Metric	Meaning	1 of 2 spans
Hit	At least one span covered	1
Evidence recall	Share of spans covered	0.5
All-evidence	Every span covered	0

Team method · Macro average over 20 questions · Gold spans are selected, not exhaustive

TF-IDF retrieved more evidence here

Macro chunk Recall@k, 192 tokens: TF-IDF (dark) · MiniLM (orange)



At k = 3

Metric	TF-IDF	MiniLM
Hit	85%	65%
Evidence	80%	62.5%
All	75%	60%

MiniLM still wins on Q06 and Q13. One paper; not a general ranking.

This motivated a follow-up experiment with paraphrased questions.

Team experiment · results/retrieval_summary.csv · n = 20 questions

Does rewording change retrieval performance?

20 paired questions: original vs. paraphrased

Same intended answers, gold spans, corpus, and retrieval settings.
Retrieval only — no additional Gemini calls.

Example: Q06

Original: “minimum Python and R versions”
Paraphrased: “oldest supported releases of Python and R”

Evidence recall (not the chunk Recall@k charted on the previous slide) · 192-token chunks, k = 3

Retriever	Original	Paraphrased	Change
TF-IDF	80%	55%	−25 percentage points
MiniLM	62.5%	55%	−7.5 percentage points

MiniLM showed a smaller decline, but both methods reached the same evidence recall after paraphrasing.

Exploratory study; one paraphrase per question. No generation evaluation on paraphrases. Source: results/paraphrase_summary.json

Worked example: from question to citation

Q08 At the time of writing, what were the Python and R package test coverages?

Retrieved “The coverages for the Python and the R packages at the time of writing are 97% and 96%, respectively.”
Top chunk s192-p2-t336, page 2

Gemini “At the time of writing, the code coverages for the Python and the R packages were 97% and 96%, respectively [s192-p2-t336].”

Checked: both numbers, their order, the time qualifier, and the cited chunk.

Retrieved context improved answer correctness

Condition	Full	Partial	Incorrect	Abstained	Mean
Closed-book	5/20	6/20	9/20	0/20	0.400
TF-IDF RAG	16/20	2/20	0/20	2/20	0.850
MiniLM RAG	14/20	3/20	0/20	3/20	0.775

Zero Incorrect labels does not mean zero unsupported claims.

Q16: context corrects an error

Closed-book says 100 test sets; both RAG answers say 20, as the paper does.

Closed-book never abstained

It was told to admit unknowns, yet still gave specific wrong values.

Gemini 3.5 Flash-Lite, 192 tokens, k = 3 · AI-assisted review accepted by both authors; not blind

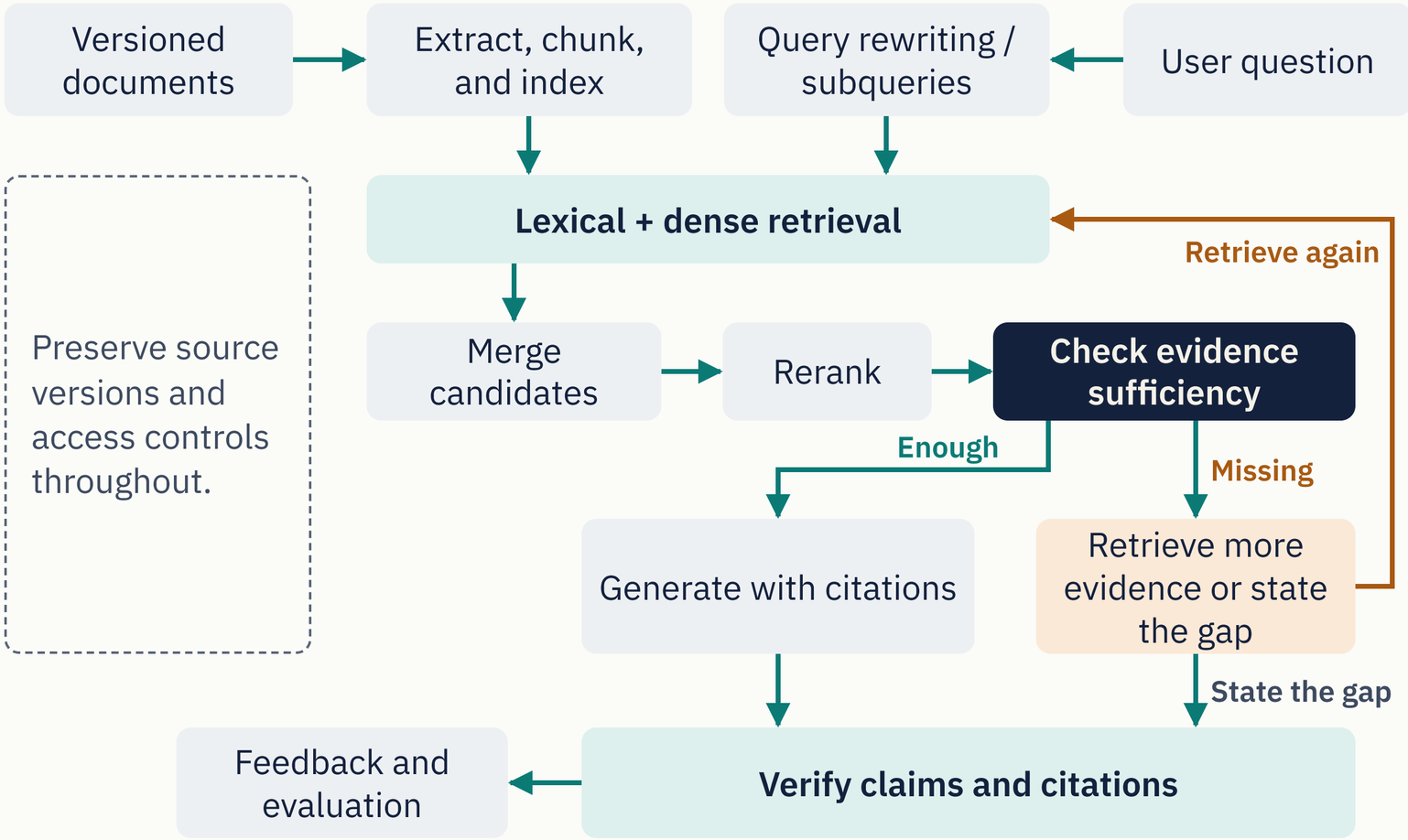
Failures we observed, by category

Category	Case	What happened
Missing retrieval	Q06 TF-IDF	Version passage absent; the model abstained
Incorrect retrieval	Q13 TF-IDF	Got the conclusion and references, not the architecture
Partial evidence	Q09 both	Each retriever found only one of two licenses
Beyond evidence	Q07 MiniLM	Inferred a service’s role from an /actions URL
Irrelevant citation	None clear	Nearest case: Q15 cites a chunk for a mistaken reason

A valid chunk ID does not prove that the chunk supports the claim.

results/generation.jsonl, results/answer_review.csv · Verifiability: Schneider, 2024

Proposed extension: an organizational RAG pipeline



From our experiment

Wording sensitivity

Query rewriting and hybrid retrieval

Related but unhelpful passages

Reranking

Missing or unsupported facts

Evidence checks and claim verification

Proposed, not implemented or evaluated. Informed by RAG-Fusion and Corrective RAG; see report Appendix B.

Documents reduced errors here, but cannot prove hallucination is gone

To rely on RAG

Check each claim against its source
Track document versions and hashes
Reindex and rerun tests on change

Limits of this study

20 answerable questions, one paper
One small embedding model, one run
Selected gold spans, no blind review

Evaluate query variation, not only original-question performance.

Lewis 2020; Schneider 2024; Zhu 2022 · AI tools: Codex, Claude Code; Gemini was the tested model